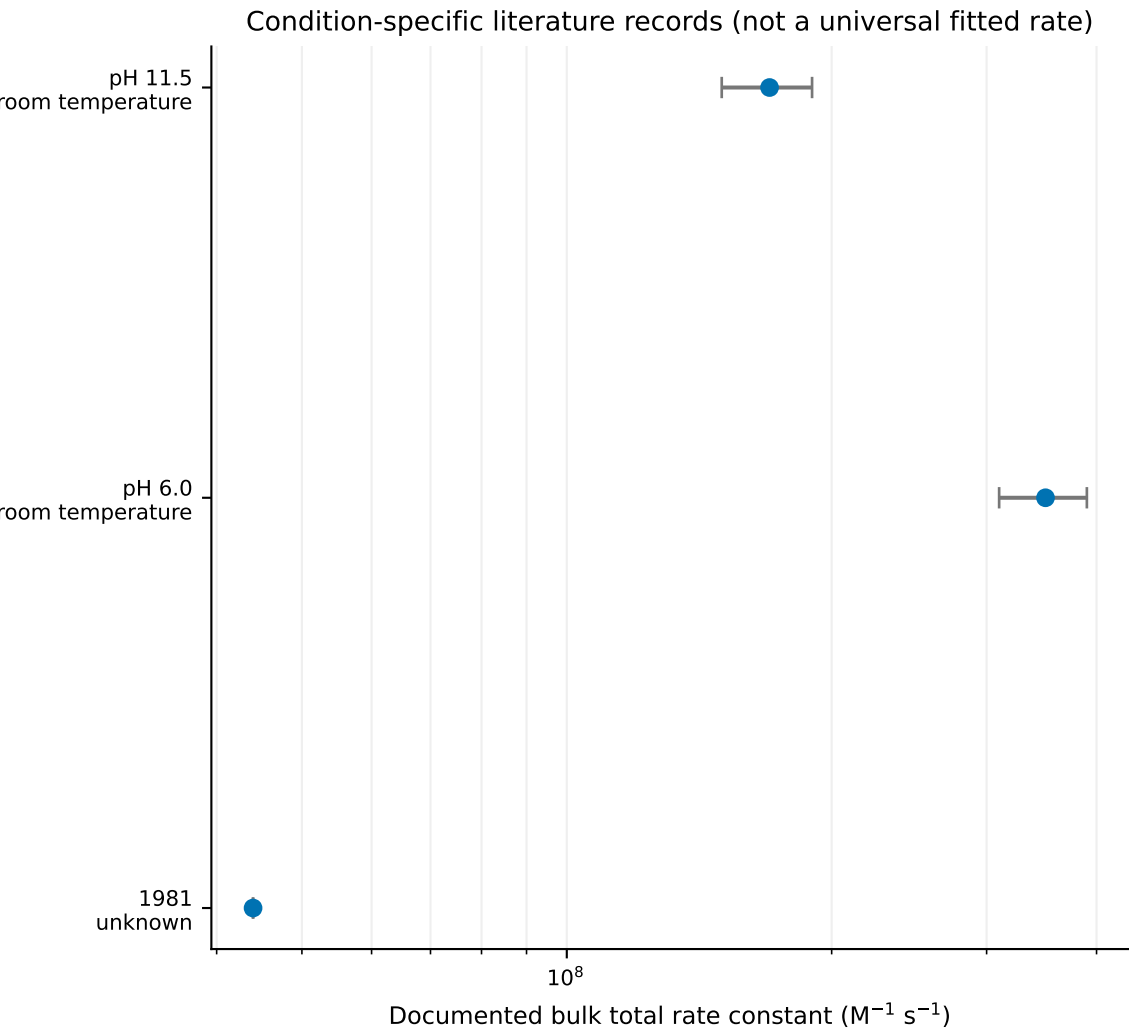




Records remain conditionally distinct

Record	Conditions/environment	Species/protonation	Source/allowed use
1981	pH unknown; unknown; aqueous pulse radiolysis	adriamycin semiquinone plus ground-state triplet O ₂ ; protonation: unknown	B. A. Svingen and G. Powis, Archives of Biochemistry and Biophysics 209(1), 119-126 (1981), doi:10.1016/0003-9861(81)90263-0; allowed: literature_arithmetic_sensitivity_only
pH6	pH 6.0; room temperature; aqueous phosphate/formate; O ₂ or O ₂ /N ₂ purged	adriamycin semiquinone plus neutral ground-state triplet O ₂ ; protonation: biological-pH internally protonated/zwiterionic semiquinone; O ₂ neutral	Land et al. (1985), doi:10.1038/bjc.1985.74; allowed: literature_arithmetic;condition_matched_prediction;sensitivity_only
pH11_5	pH 11.5; room temperature; aqueous borate/OH ⁻ ; O ₂ or O ₂ /N ₂ purged	deprotonated adriamycin semiquinone plus neutral ground-state triplet O ₂ ; protonation: deprotonated semiquinone; O ₂ neutral	Land et al. (1985), doi:10.1038/bjc.1985.74; allowed: literature_arithmetic;condition_matched_prediction;sensitivity_only